



PROJECT REPORT

HEALTHCARE APPOINTMENT MANAGEMENT SYSTEM

Submitted By

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Submitted To

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CERTIFICATE

This is to certify that Himanshi, student of Mr. Kannaiya Raja N, has successfully completed the project titled “**Healthcare Appointment Management System**” as part of the academic requirements for the subject Introduction To Problem Solving during the academic year **2026-27**.

The project has been completed under my guidance and supervision.

Teacher's Signature: _____

Date: 30-09-2026

Place: VIT Bhopal University

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my teacher/professor for providing me with the opportunity to work on the project

“Healthcare Appointment Management System.”

I am thankful for the guidance, support, and valuable suggestions provided throughout the development of this project.

I would also like to thank my institution, classmates, friends, and family for their encouragement and support during the completion of this project.

Finally, I am grateful for the opportunity to improve my programming knowledge and practical understanding of Python through this project.

ABSTRACT

The **Healthcare Appointment Management System** is a simple console-based application developed using the Python programming language.

The main purpose of this project is to demonstrate how a basic hospital appointment system can be implemented using programming concepts such as user input, variables, conditional statements, and nested decision-making.

The system collects basic patient information such as name, age, and mobile number. It then allows the patient to select a hospital department. Currently, the system provides functionality for the Clinical Department, where the patient can choose a medical specialty such as Cardiology, Neurology, Orthopedics, or Paediatrics.

After selecting a specialty, the patient can choose a doctor. The system displays the doctor's qualifications, experience, and available consultation hours before confirming the appointment.

This project is designed primarily for educational purposes and demonstrates the basic working principles of a hospital appointment management application.

1. INTRODUCTION

Healthcare management systems are widely used to organize patient information, doctor details, appointments, and hospital services.

Manually managing appointments can require considerable effort and may result in errors. A computerized appointment system can simplify the process by allowing patients to select departments and doctors through an organized interface.

The **Doctor Appointment Management System** developed in this project is a basic Python application that demonstrates this concept through a command-line interface.

The application uses a series of menus and conditional statements to guide the user through the appointment selection process.

2. PROBLEM STATEMENT

In a traditional manual appointment system, patients may need to communicate with hospital staff to determine:

- Which department they should visit.
- Which doctors are available.
- What qualifications a doctor has.
- When the doctor is available.
- Whether their appointment has been successfully registered.

A simple computerized system can organize this information and provide it to the user in a structured way.

Therefore, this project aims to develop a basic **Doctor Appointment Management System** using Python.

3. OBJECTIVES OF THE PROJECT

The main objectives of this project are:

1. To develop a simple hospital appointment system using Python.
 2. To collect basic patient information.
 3. To provide department selection.
 4. To provide different clinical specialties.
 5. To allow users to select a doctor.
 6. To display doctor qualifications and experience.
 7. To display doctor availability.
 8. To provide appointment confirmation.
 9. To demonstrate the use of conditional statements in Python.
 10. To develop practical programming skills.
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4. SCOPE OF THE PROJECT

The current project focuses on basic appointment selection through a console-based application.

The system includes:

- Patient registration through basic input.
- Department selection.
- Clinical department selection.
- Doctor selection.
- Doctor information.
- Doctor availability.
- Appointment confirmation.

The current version is intended as a basic prototype and does not maintain permanent patient or appointment records.

5. TECHNOLOGIES USED

Programming Language

Python 3

Python was selected because it is easy to learn, readable, and suitable for developing beginner-level applications.

Concepts Used

The project uses the following Python concepts:

- Variables
- Data types
- **input()**
- **print()**
- **if** statements
- **elif** statements
- **else** statements
- Nested conditional statements
- String handling
- Basic user interaction

Hardware Requirements

A basic computer or laptop is sufficient.

Recommended:

- Processor: Dual-core or better
- RAM: 2 GB or more
- Storage: At least 100 MB of free space

Software Requirements

- Python 3.x
- Any Python-compatible IDE or code editor

Examples include:

- IDLE
 - Visual Studio Code
 - PyCharm
 - Jupyter-compatible environments
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6. SYSTEM REQUIREMENTS

Input Requirements

The program accepts:

- Patient name
- Patient age
- Mobile number
- Hospital department
- Clinical specialty
- Doctor selection

Output Requirements

The program displays:

- Welcome message
 - Selected department
 - Selected doctor
 - Doctor qualification
 - Doctor experience
 - Doctor availability
 - Appointment confirmation
-

7. SYSTEM DESIGN

The application follows a simple hierarchical decision-making structure.

START

|

V

Enter Patient Details

|

+--> Name

+--> Age

+--> Mobile Number

|

V

Select Department

|

+--> Clinical

|

+--> Diagnostic

|

+--> Administrative

|

V

If Clinical

|

V

Select Clinical Specialty

|

+--> Cardiology

+--> Neurology

+--> Orthopedics

+--> Paediatrics

|

V

Select Doctor

|

V

Display Doctor Information

|

V

Confirm Appointment

|

V

END

8. MODULE DESCRIPTION

8.1 Patient Information Module

The first section of the program collects the patient's basic information.

```
name = input("Enter your name: ")
```

```
age = int(input("Enter your age: "))
```

```
Mobile_number = int(input("Enter your mobile number: "))
```

The information is stored in variables and used during the interaction.

8.2 Department Selection Module

The user is asked to select a hospital department.

```
department = input(
```

```
    "Enter the department you want to visit : "
```

```
    "(Clinical=1, diagnostic=2, administrative=3): "
```

```
)
```

The program then checks the user's selection using conditional statements.

8.3 Clinical Department Module

If the user selects Clinical Department, the program displays several specialties.

These include:

- Cardiology
- Neurology
- Orthopedics
- Paediatrics

The program also lists additional specialties in the menu, although they are not yet implemented in the current code.

8.4 Doctor Selection Module

After selecting a specialty, the user can choose one of the available doctors.

For example, the Cardiology department provides:

- Dr. Satish
- Dr. Ravi
- Dr. Rakesh

The selected doctor's information is then displayed.

8.5 Doctor Information Module

The program displays information such as:

- Doctor's name
- Qualifications
- Experience
- Available consultation hours

For example:

You have selected Dr. Satish.

Qualifications: MBBS, MD (Cardiology)

Experience: 15 years in Cardiology

Dr. Satish is available from 10:00 AM to 2:00 PM.

8.6 Appointment Confirmation Module

After displaying the doctor's information, the program informs the patient that the appointment has been successfully booked.

It also instructs the patient to arrive 15 minutes before the scheduled time.

9. DEPARTMENT AND DOCTOR DETAILS

Cardiology

Doctor	Qualification	Experience	Availability
Dr. Satish	MBBS, MD (Cardiology)	15 years	10 AM - 2 PM
Dr. Ravi	MBBS, MD (Cardiology)	10 years	2 PM - 6 PM
Dr. Rakesh	MBBS, MD (Cardiology)	8 years	6 PM - 10 PM

Neurology

Doctor	Qualification	Experience	Availability
Dr. Sara	MBBS, MD (Neurology)	12 years	10 AM - 2 PM
Dr. Sakshi	MBBS, MD (Neurology)	9 years	2 PM - 6 PM
Dr. Samay	MBBS, MD (Neurology)	6 years	6 PM - 10 PM

Orthopedics

Doctor	Qualification	Experience	Availability
Dr. Ramesh	MBBS, MS (Orthopedics)	14 years	10 AM - 2 PM
Dr. Kavita	MBBS, MS (Orthopedics)	11 years	2 PM - 6 PM
Dr. Manoj	MBBS, MS (Orthopedics)	9 years	6 PM - 10 PM

Paediatrics

Doctor	Qualification	Experience	Availability
Dr. Ananya	MBBS, MD (Paediatrics)	13 years	10 AM - 2 PM
Dr. Ritu	MBBS, MD (Paediatrics)	10 years	2 PM - 6 PM
Dr. Karan	MBBS, MD (Paediatrics)	7 years	6 PM - 10 PM

10. ALGORITHM

The algorithm of the system is as follows:

1. Start the program.
 2. Ask the user to enter their name.
 3. Ask the user to enter their age.
 4. Ask the user to enter their mobile number.
 5. Display the hospital welcome message.
 6. Ask the user to select a department.
 7. If Clinical Department is selected:
 - Display clinical specialties.
 - Ask the user to select a specialty.
 8. Check the selected specialty.
 9. Display available doctors for that specialty.
 10. Ask the user to select a doctor.
 11. Display the selected doctor's details.
 12. Display doctor availability.
 13. Display appointment confirmation.
 14. If an unsupported department is selected, display an appropriate message.
 15. End the program.
-

11. PSEUDOCODE

START

INPUT name

INPUT age

INPUT mobile number

DISPLAY welcome message

INPUT department

IF department = Clinical THEN

 DISPLAY clinical departments

 INPUT clinical department

 IF Cardiology THEN

 DISPLAY cardiology doctors

 INPUT doctor

 DISPLAY selected doctor's details

 DISPLAY appointment confirmation

 ELSE IF Neurology THEN

 DISPLAY neurology doctors

 INPUT doctor

 DISPLAY selected doctor's details

 DISPLAY appointment confirmation

ELSE IF Orthopedics THEN

 DISPLAY orthopedics doctors

 INPUT doctor

 DISPLAY selected doctor's details

 DISPLAY appointment confirmation

ELSE IF Paediatrics THEN

 DISPLAY paediatrics doctors

 INPUT doctor

 DISPLAY selected doctor's details

 DISPLAY appointment confirmation

ELSE

 DISPLAY invalid selection

ELSE

 DISPLAY department not configured

END

12. SAMPLE OUTPUT

Enter your name: Rahul

Enter your age: 25

Enter your mobile number: 9876543210

Hi Rahul, Welcome to Cygnus Hospital.

Enter the department you want to visit:

(Clinical=1, diagnostic=2, administrative=3): 1

You have selected Clinical Department.

Enter the specific clinical department you want to visit:

Cardiology=1, Neurology=2, Orthopedics=3, Paediatrics=4,

Oncology (cancer)=5, Obstetrics & Gynaecology=6: 1

You have selected Cardiology Department.

Enter the doctor you want to visit:

(Dr. Satish=1, Dr. Ravi=2, Dr. Rakesh=3): 1

You have selected Dr. Satish.

Qualifications: MBBS, MD (Cardiology)

Experience: 15 years in Cardiology

Dr. Satish is available from 10:00 AM to 2:00 PM.

Your appointment with Dr. Satish has been successfully booked.

Please arrive 15 minutes before your scheduled time.

Thank you for using the Healthcare Management System.

We hope you have a great experience at Cygnus Hospital.

13. ADVANTAGES

The system provides several advantages:

- Simple and easy to use.
 - Beginner-friendly interface.
 - Reduces manual selection steps.
 - Provides organized doctor information.
 - Demonstrates basic automation.
 - Easy to modify and expand.
 - Helps beginners understand Python programming concepts.
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14. LIMITATIONS

The current version has some limitations:

1. It does not store patient information permanently.
 2. It does not use a database.
 3. Appointment dates cannot currently be selected.
 4. The user cannot select an exact appointment time.
 5. Appointment cancellation is not available.
 6. Appointment rescheduling is not available.
 7. Diagnostic Department is not implemented.
 8. Administrative Department is not implemented.
 9. Oncology and Obstetrics & Gynaecology are listed but not implemented.
 10. Input validation is limited.
 11. The program does not have a graphical user interface.
 12. It does not verify real-time doctor availability.
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15. FUTURE ENHANCEMENTS

The system can be improved significantly in future versions.

Database Integration

A database such as SQLite or MySQL can be added to store:

- Patient records
- Doctor records
- Appointments
- Department information

Appointment Scheduling

Users could select:

- Appointment date
- Appointment time
- Available time slots

Appointment Management

The system could provide options to:

- Book appointments
- Cancel appointments
- Reschedule appointments
- View existing appointments

Graphical User Interface

A GUI could be developed using Python's Tkinter library to make the application easier to use.

Web Application

The system could later be converted into a web application using frameworks such as Flask or Django.

Authentication

Patient and administrator login systems could be introduced.

16. TESTING

Testing is performed to ensure that the program responds correctly to different user inputs.

Test Case	Input	Expected Result
1	Clinical = 1	Clinical Department is selected
2	Cardiology = 1	Cardiology Department is selected
3	Doctor = 1	Dr. Satish information is displayed
4	Doctor = 2	Dr. Ravi information is displayed
5	Doctor = 3	Dr. Rakesh information is displayed
6	Invalid clinical option	Invalid selection message
7	Diagnostic = 2	Department not configured message
8	Administrative = 3	Department not configured message

Testing helps identify incorrect inputs and ensures that the appropriate output is generated.

17. CONCLUSION

The **Healthcare Appointment Management System** successfully demonstrates how Python can be used to create a simple healthcare appointment application.

The project uses basic programming concepts such as variables, user input, conditional statements, and nested decision-making to create an interactive console application.

The system allows a user to enter personal details, select a department, choose a medical specialty, select a doctor, view doctor information, and receive an appointment confirmation.

Although the current project is a basic prototype, it provides a foundation for developing a more advanced hospital management system. Features such as databases, appointment scheduling, authentication, cancellation, rescheduling, and graphical interfaces can be added in future versions.

Overall, this project provides practical experience in Python programming and demonstrates how programming concepts can be applied to a real-world problem.

18. REFERENCES

The following resources can be used for learning and developing this project:

1. Python programming documentation and tutorials.
2. Python 3 programming concepts.
3. Basic concepts of hospital management systems.
4. Classroom notes and programming exercises.
5. Python **input()** and conditional statement concepts.

19. SOURCE CODE

The main source code for this project is stored in:

`Healthcare_appointment_management_system.py`

The project documentation is stored in:

`README.md`

20.PROJECT SUMMARY

Project Title: Doctor Appointment Management System

Programming Language: Python

Project Type: Console-Based Application

Domain: Healthcare / Hospital Management

Main Concept: Doctor Appointment Selection

Database: Not currently used

Interface: Command Line / Console

Target Users: Patients / Demonstration Users

Purpose: Educational Project